

Programming for Analytics

- RDBMS and SQL

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ACCREDITATIONS



MEMBERSHIPS



RANKINGS



Overview of the class

Discipline "Programming for Analytics"

Motivation

Relational Database

Relational Database Management Systems (RDMS)

Structure Query Language (SQL)

Subject

- SQL (2 classes, 3 hours each): Introduction to SQL, including fundamental concepts and practical usage.
- Python (3 classes, 3 hours each): Covers Python basics, algorithm design, and visualization techniques.

By the end of the course, students will be able to develop Python and SQL-based applications for AI projects.

Software

SQL – MySQL Workbench

Python – Anaconda (Jupyter Notebook)

ECTS

ECTS: 3

Lective Hours: 13.5 hours

Evaluation Hours: in-class evaluation: 1.5 hours

Evaluation

Evaluation

Component	Weight	Details
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Quizzes	20%	One per session (sessions 1–4), administered at the end of class (last 10 minutes) via In Virtual Campus. 5 points each.
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Group Project	40%	Continuous project developed across all sessions. Final deliverables submitted after Session 5: code and slide presentation. Mandatory.
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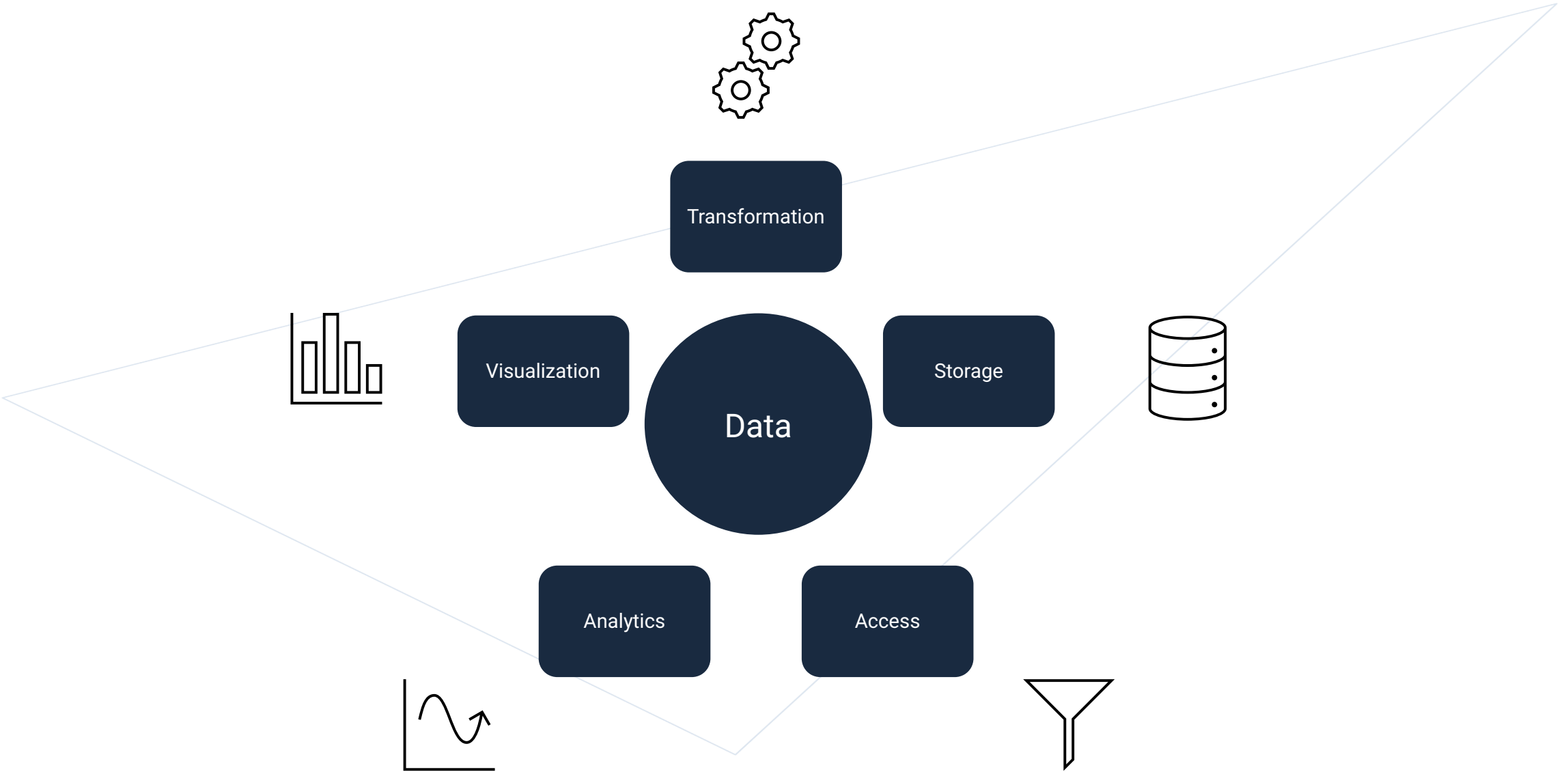
Exam	40%	Covers all course content: 50% SQL / 50% Python. Held in the last hour of Session 5. Mandatory.
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Important Rules

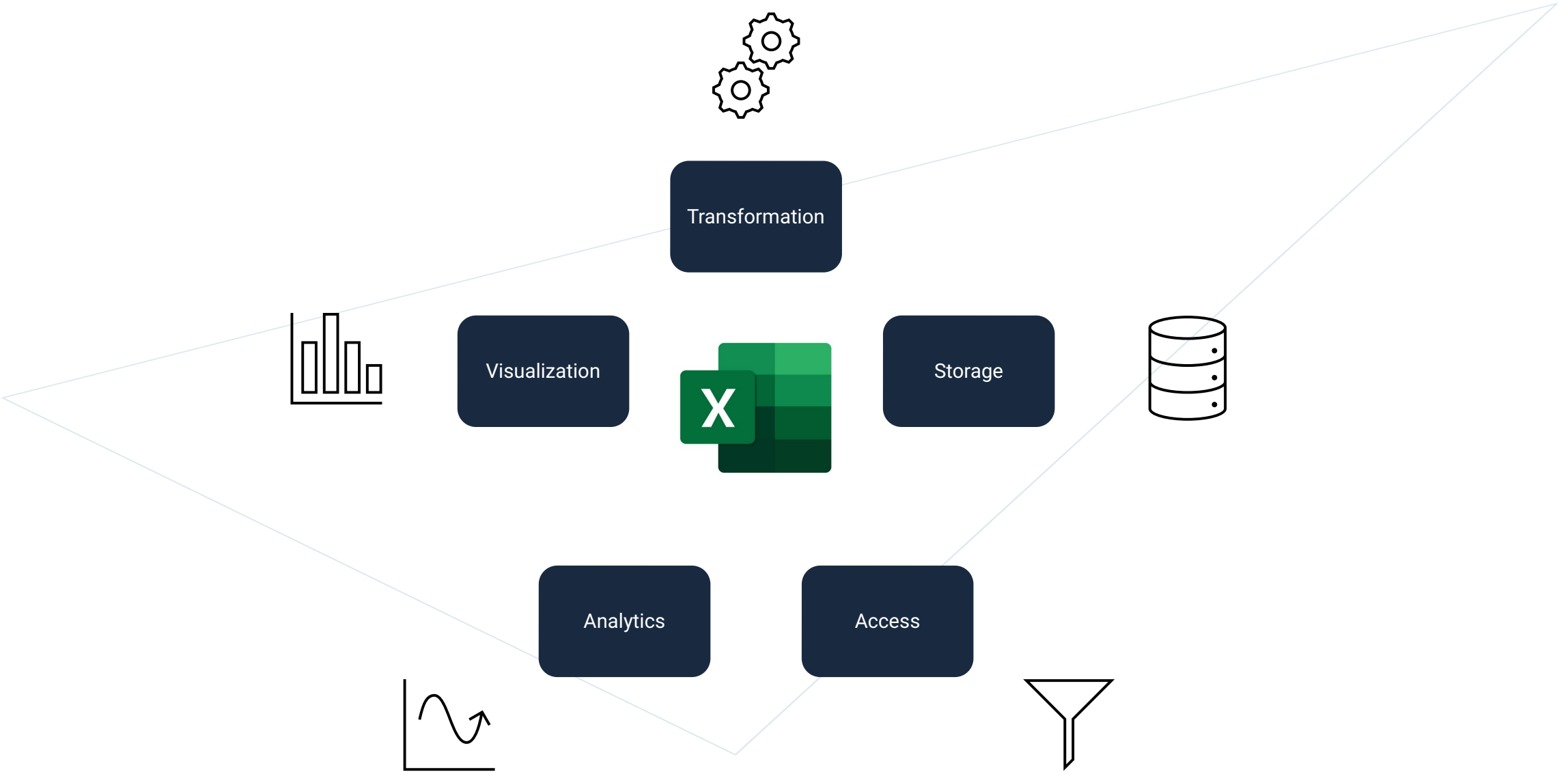
- Both the **project** and the **exam** are mandatory components. Failure to submit or sit either will result in course failure.
- Students who pass in the normal evaluation period are **not eligible** for the extraordinary evaluation period.
- **Extraordinary evaluation:** exam only (40% of the grade). Neither the project (40%) nor the quizzes (20%) can be recovered through extraordinary evaluation.

Quizzes and mini-test are individual and you cannot use generative models (like Chatgpt, Perplexity, Copilot or others)

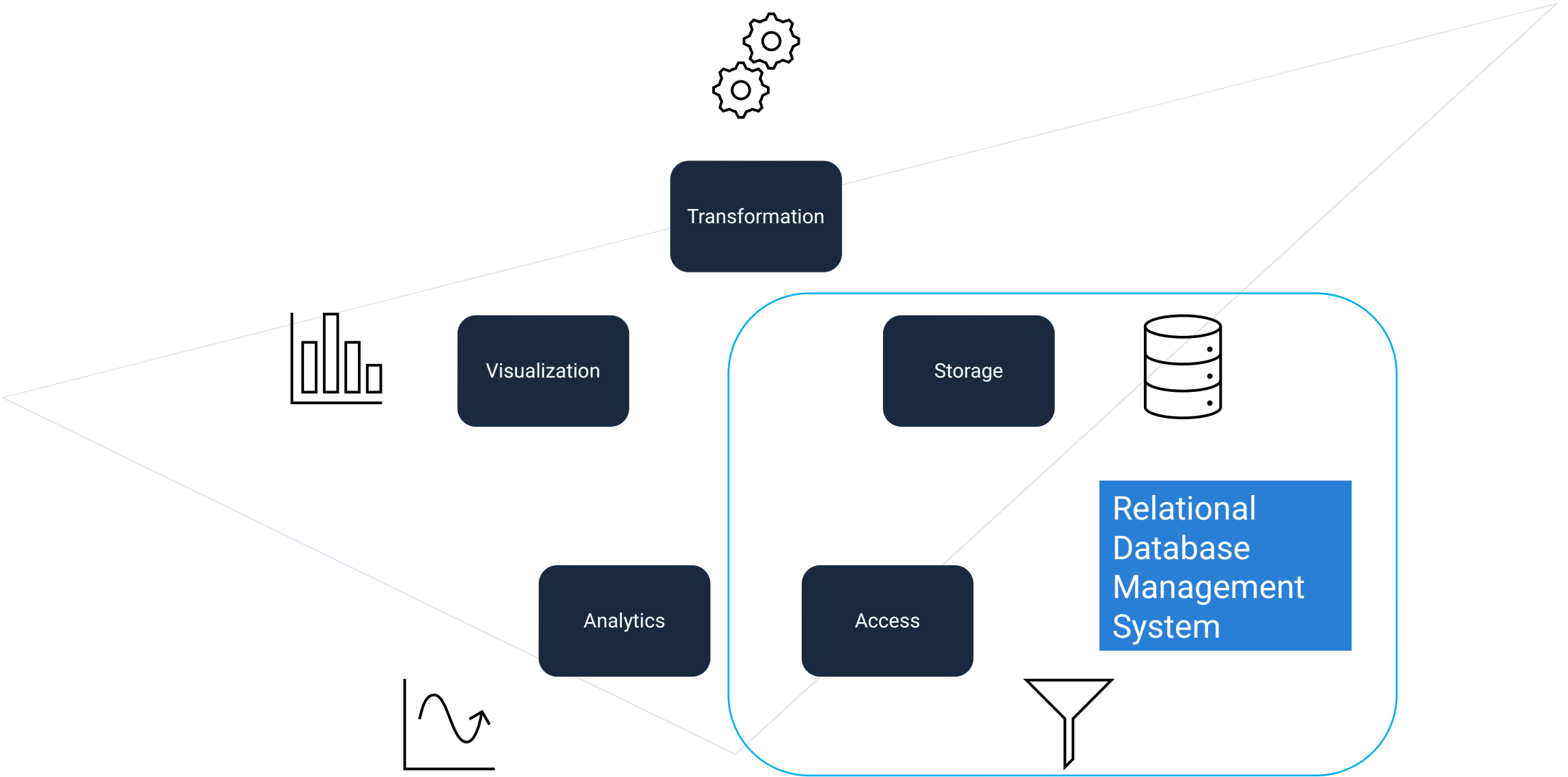
Motivation



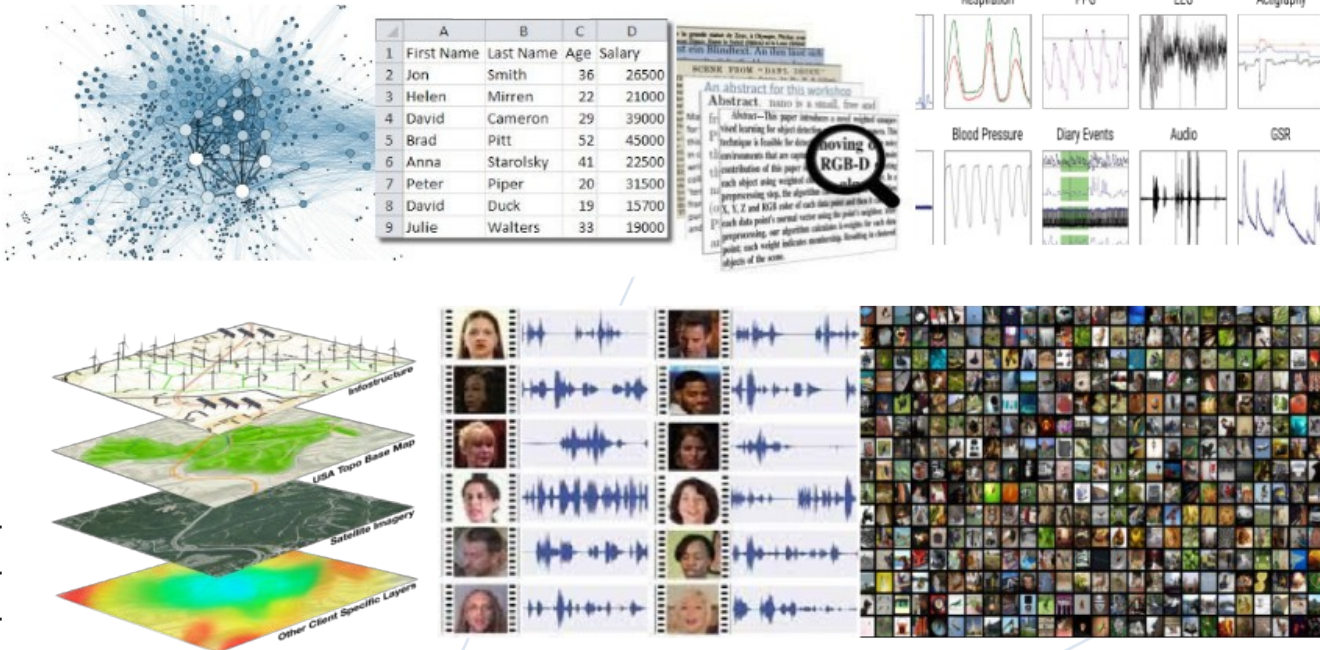
Motivation



Motivation



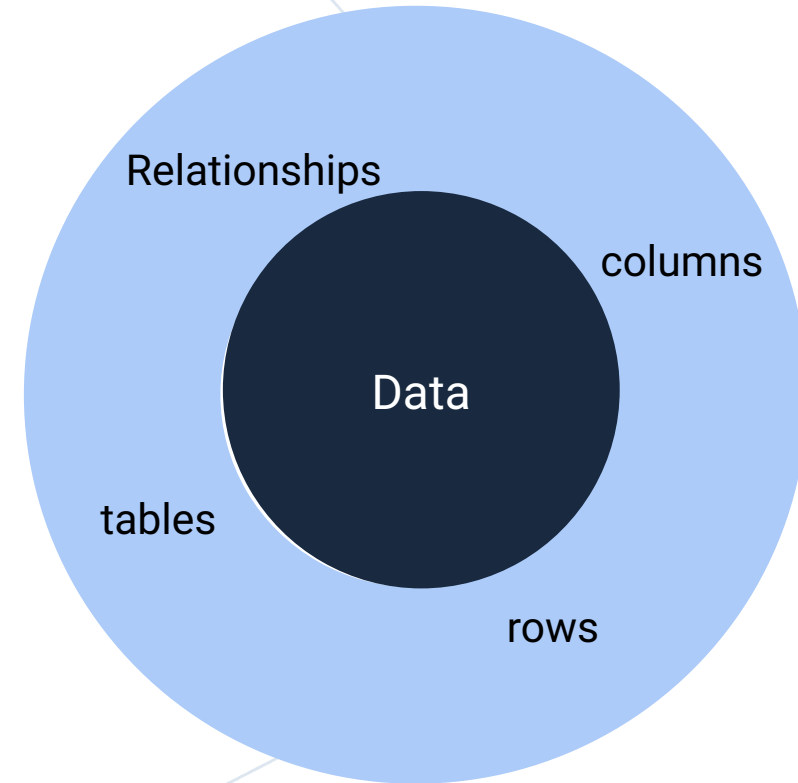
Relation Database Management System (RDBMS) mainly deals with **structured data**, but modern RDBMS can also handle **semi-structured data** like JSON, XML, and even spatial data. However, it is not ideal for **unstructured data** like raw video, audio, or logs, which are better suited for NoSQL databases.



Relational Database – What is it?

A relational database is a collection of information that organizes data in predefined relationships where

- Data is stored in **one or more tables** of **columns** and **rows**, making it easy to see and understand how different data structures relate to each other.
- A *table* is a set of data elements (values) using a model of vertical columns (identifiable by name) and horizontal rows, the cell being the unit where a row and column intersect
- Relationships are a logical connection between different tables, established on the basis of interaction among these tables.



Relationship

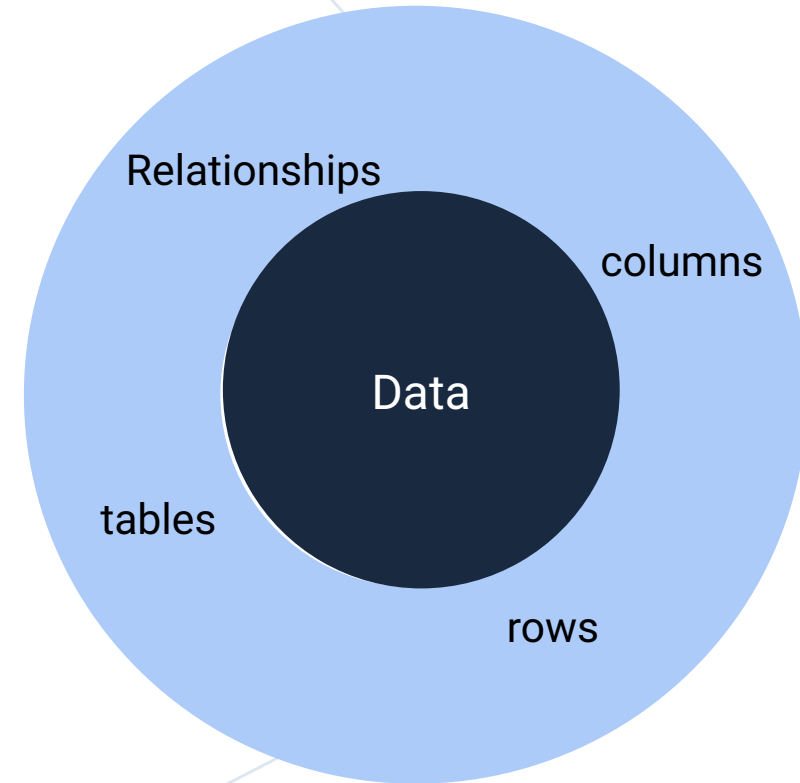
Consider the following two tables:

1. Students Table

StudentID	Name	Age	Class
1	John	15	10A
2	Mary	14	9B
3	Alex	16	11A

2. Courses Table

CourseID	CourseName	StudentID
101	Mathematics	1
102	Science	2
103	English	1



Relationship Type: One-to-Many

A student can be enrolled in multiple courses.

The StudentID in the Courses table links to the Students table.

Relationship

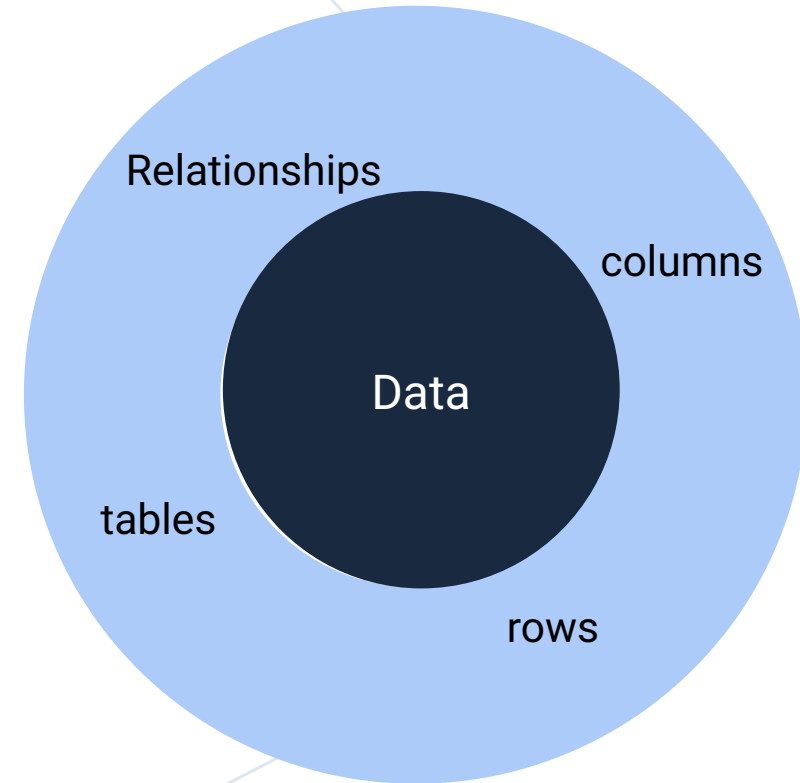
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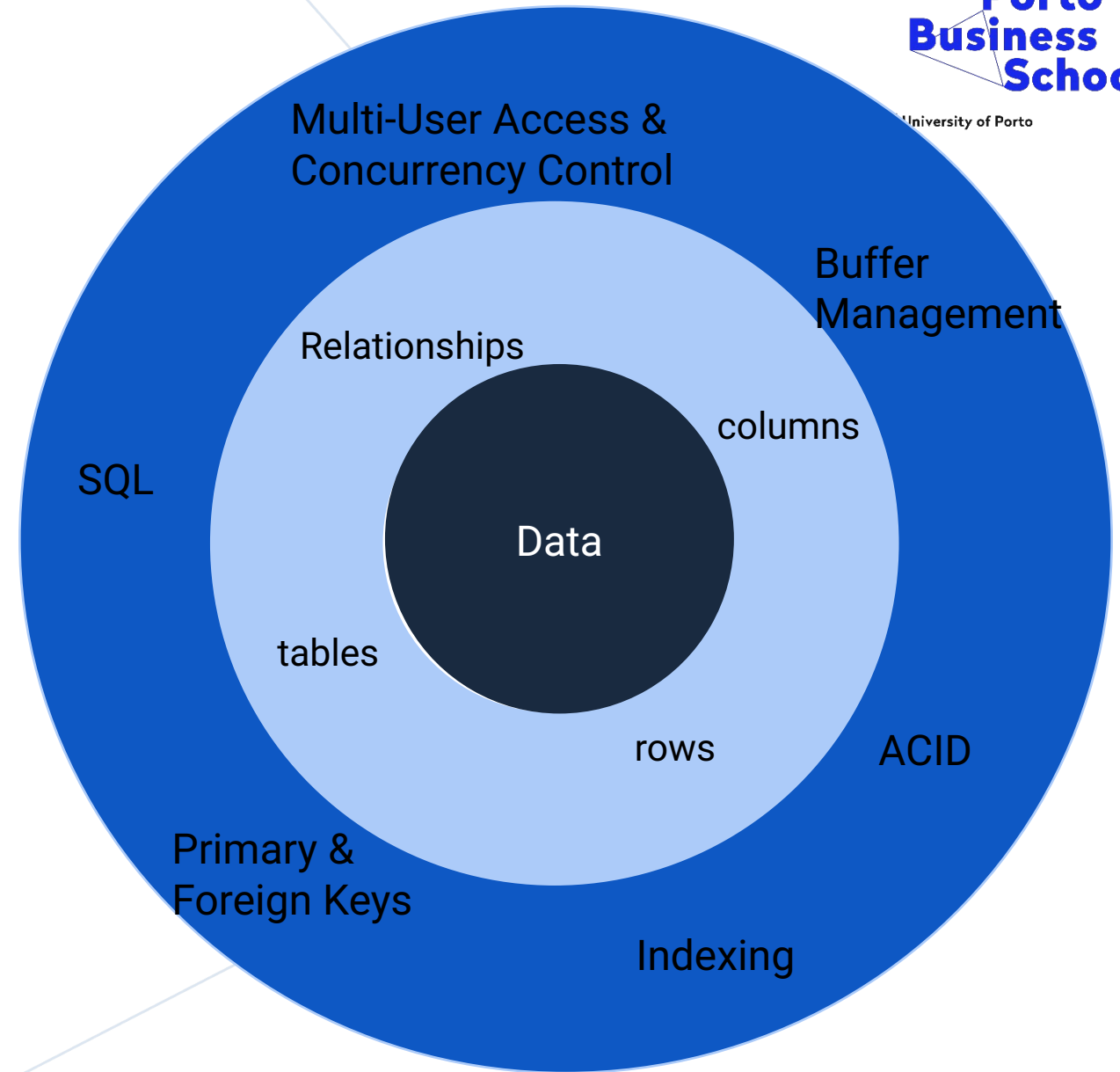
The primary key is a unique identifier within its table, whereas a foreign key is a reference in one table to a primary key in another.

Which is the PK and FK?

Relational DB Management System

Now that we have a relational database, we can use a Relational Database Management System (RDBMS) to create, update, and manage.

With database well-defined (tables, columns, relationships, etc), RBMS offer multiple tools and processes gives advantages before others tools.



Advantage of using RDBMS

- **Powerful Querying with SQL**
 - With a simple language, I can access, add or delete only the data that you need
- **Reproducibility**
 - SQL queries can be saved and reused
 - Pipelines become more transparent and repeatable
 - Easier to audit how data was transformed
- **Powerful Querying with SQL**
 - With a simple language, I can access, add or delete only the data that you ne
- **Data Integrity & Accuracy**
 - Relational databases enforce:
 - Constraints (e.g., primary keys, foreign keys)
 - Validation rules
 - This ensures high-quality, reliable data—critical for building accurate models.

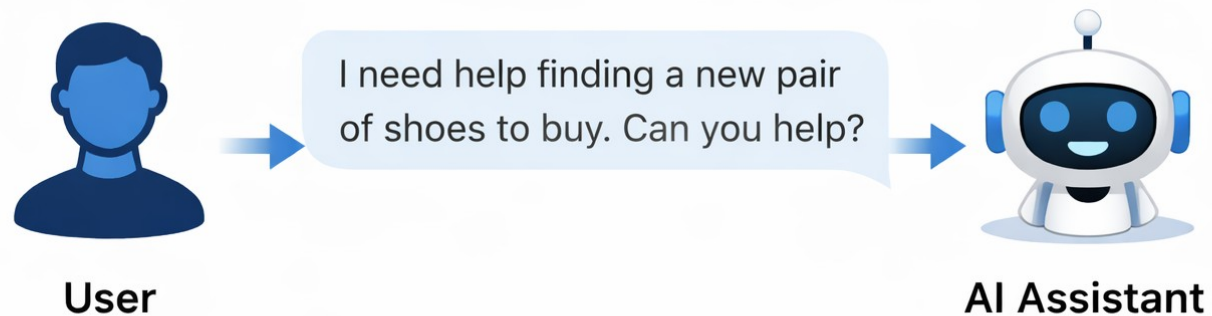


Advantage of using RDBMS

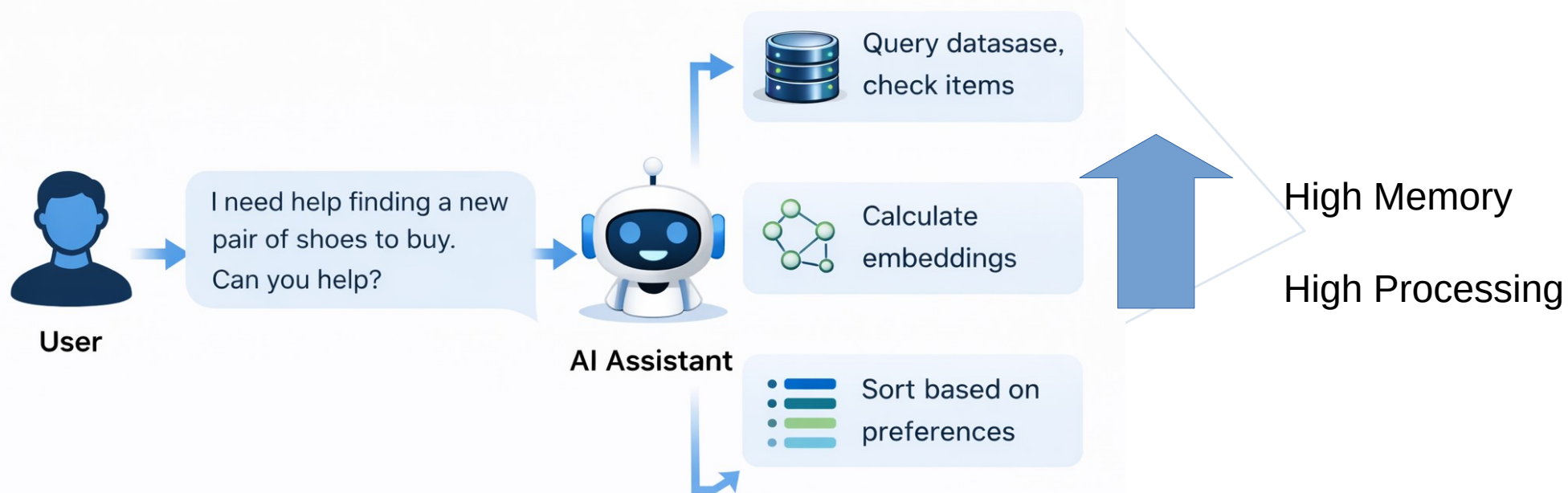
- **Structured & Organized Data**
 - Define tables, columns, data type, relationships, etc
- **Efficient Handling of Large Data**
 - Using SQL, just use load data that you need
 - Using some functionalities like indexes, partitions, etc can optimize for fast access
- **Scalability & Collaboration**



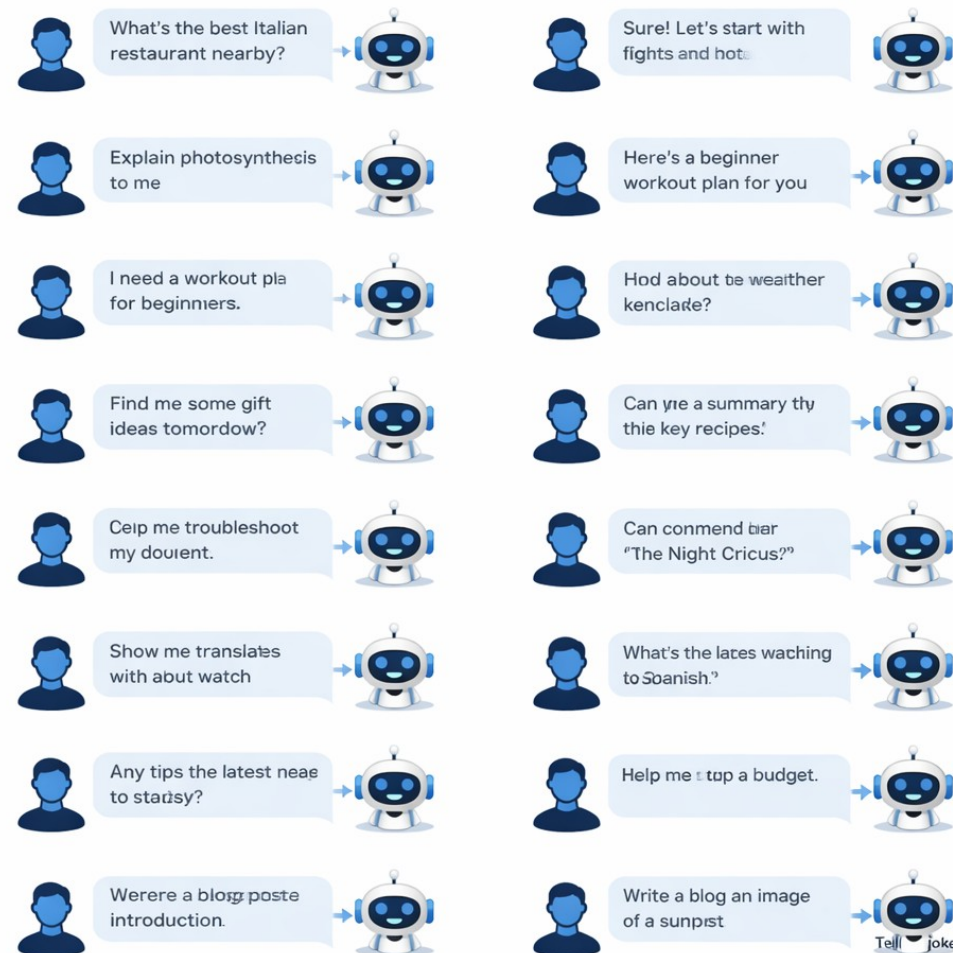
Advantage of using RDBMS – My case



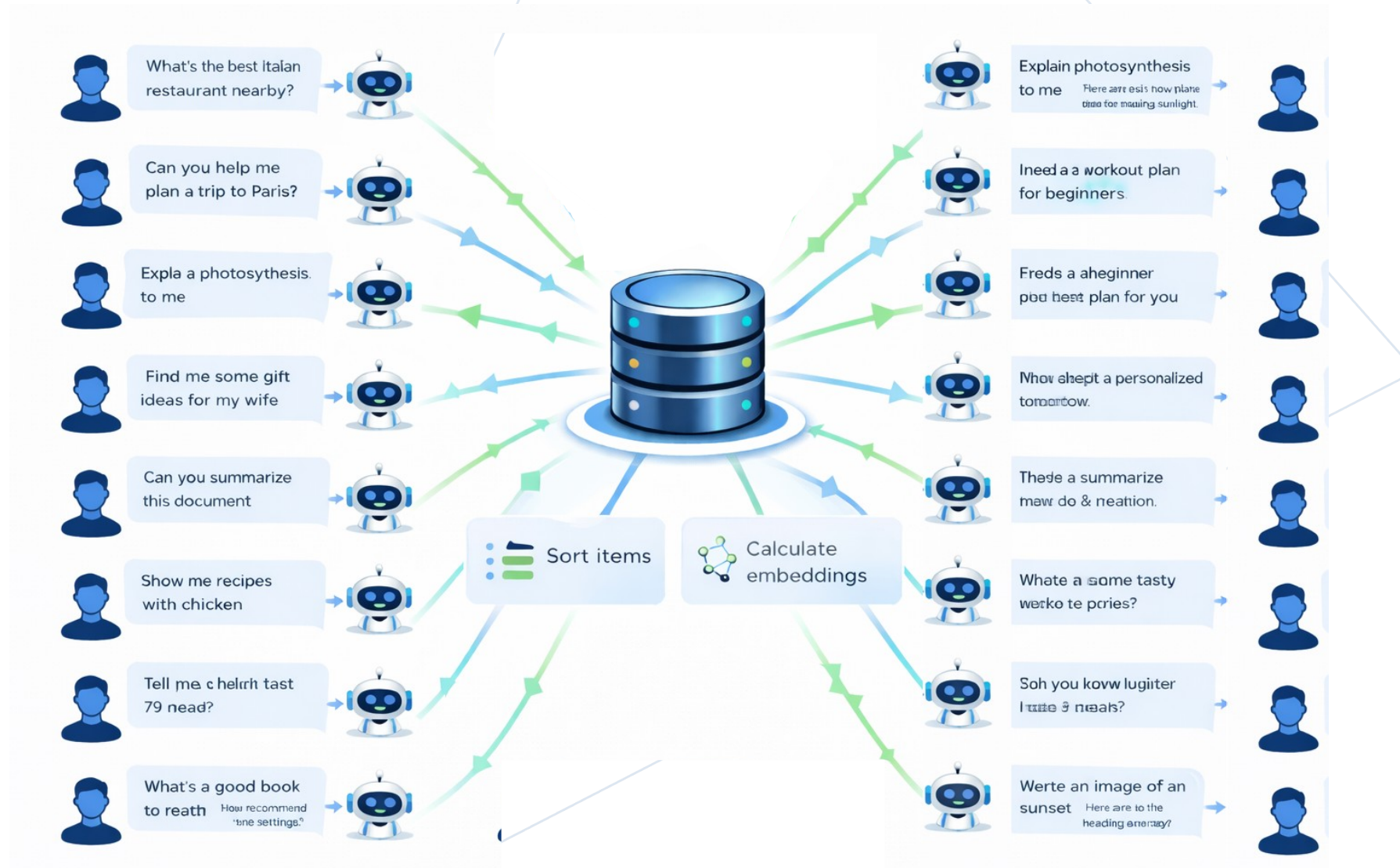
Advantage of using RDBMS – My case



Advantage of using RDBMS – My case



Advantage of using RDBMS – My case



What is SQL?

SQL (Structured Query Language) is a standard language used to store, retrieve, and manipulate data in relational databases.

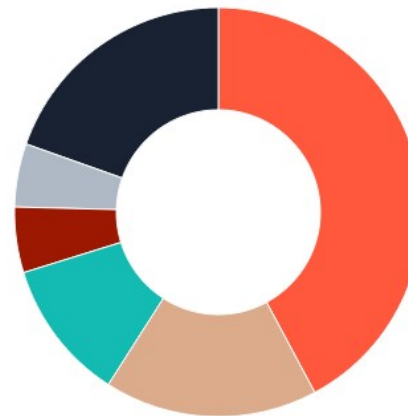
It allows end-users to communicate with databases and perform tasks like **creating, updating, and deleting databases**.



Several vendors provide relational database management systems (RDBMS), including Oracle, Microsoft, IBM, and open-source solutions like MySQL and PostgreSQL. **Beware that SQL syntax may vary slightly between these vendors**

Top 5 Relational Databases technologies in 2025

Over 265,881 companies are using Relational Databases tools. MySQL with 42.16% market share (112,092 customers), PostgreSQL with 16.91% market share (44,959 customers), Oracle Database with 11.16% market share (29,666 customers).



Technology	Market Share (Est.)	Customers
MySQL	42.16%	112,092
PostgreSQL	16.91%	44,959
Oracle Database	11.16%	29,666
Amazon Relational Database Service (RDS)	5.18%	13,767
DB2	5.00%	13,286
Others	19.60%	52,111

<https://6sense.com/tech/relational-databases>

MySQL and MySQL Workbench



MySQL is an open-source relational database management system (RDBMS)

Official - <https://www.mysql.com/>

Learn - <https://www.w3schools.com/sql/>



MySQL Workbench is a visual database design tool that integrates SQL development, administration, database design, creation and maintenance into a single integrated development environment for the MySQL database system.

Client



Query



Data



Database Server



MySQL Workbench

1

Scripts to write queries

2

Results of queries

3

Schemas, tables and elements of a database

The screenshot displays the MySQL Workbench interface with three main components highlighted by numbered blue circles:

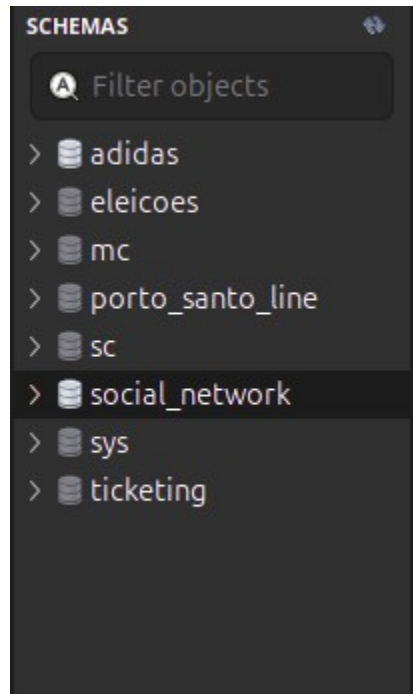
- Circle 1:** Points to the **Query Editor** window, which contains the SQL script: `1 • SELECT * FROM classicmodels.Customers;`
- Circle 2:** Points to the **Results** window, which shows a table of customer data with columns: #, customerNumbe, customerName, contactLastNam, contactFirstNam, phone, addressLine1, addressLine1, city, state, po. The table contains 14 rows of data.
- Circle 3:** Points to the **Schemas** pane on the left, which shows the database structure including schemas like classicmodels, pbs2025bag0u30, pbs2025bag0u31, etc.

The interface also includes a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help), a toolbar, and a right-hand pane with Context Help and Snippets.

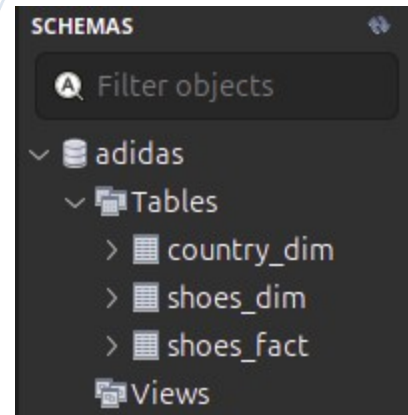
#	customerNumbe	customerName	contactLastNam	contactFirstNam	phone	addressLine1	addressLine1	city	state	po
1	103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale		Nantes	FR	440
2	112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.		Las Vegas	NV	830
3	114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	300
4	119	La Rochelle Gifts	Labrun	Janine	40.67.8555	67, rue des Cinquante Otages		Nantes	FR	440
5	121	Baane Mini Imports	Bergulfsen	Jonas	07-98 9555	Erling Skakkes gate 78		Stavern	NO	411
6	124	Mini Gifts Distributors Ltd.	Nelson	Sus	051450	5677 Strong St.		San Rafael	CA	975
7	125	Havel & Zbyszek Co	Piestrzeniewicz	Z	07-7555	ul. Filtrowa 68		Warszawa	PL	01-4
8	128	Blauer See Auto, Co.	Keitel		090 2555	Lyonerstr. 34		Frankfurt	FR	605
9	129	Mini Wheels Co.	Murphy		07	5557 North Pendale Street		San Francisco	CA	942
10	131	Land of Toys Inc.	Lee		04 44	897 Long Airport Avenue		NYC	NY	100
11	141	Euro+ Shopping Channel	Freyre		03555	C/ Moralzarzal, 86		Madrid	ES	280
12	144	Volvo Model Replicas, Co	Berglund	Cy	03555	Berguvsv		Lulea	S-9	
13	145	Danish Wholesale Impo...	Petersen	Jytte	03555	Vinb		Kobenhavn	DK	173
14	146	Savellev & Henriette Co	Savellev	Mary	03555	2, rue du Commerce		Lyon	FR	690

Exploring database/schemas

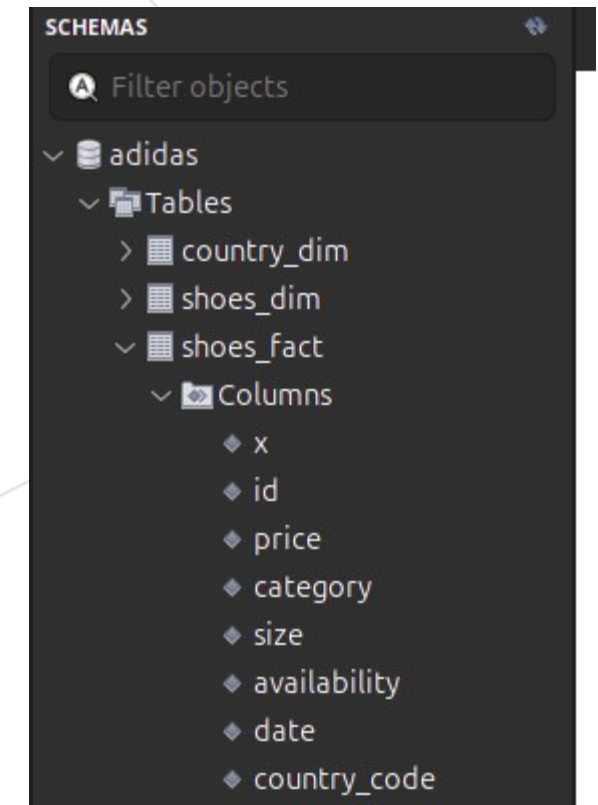
SHOW SCHEMAS;



SHOW TABLES FROM adidas;



SHOW COLUMNS FROM
adidas.shoes_fact;



SQL (MySQL) - SELECT

Get row filter by some value (WHERE) from specific columns (SELECT) from a specific table (FROM). Return the results in order by one or more columns (ORDER BY).

The WHERE and ORDER BY aren't mandatory.

Example (go to MySQL Workbench)

- "I want a table of surname, name and age of users with more than 25 years, order by age"

```
SELECT surname, name, age
FROM social_network.user_table
WHERE age >25
ORDER BY age;
```

SELECT

column1, column2, ...

Columns to retrieve

FROM

table_name

Table to query

WHERE

condition

Filter criteria

Not mandatory

ORDER BY

column3 [ASC|DESC]

Sorting order

SQL (MySQL) - Operators/Functions

Calculate a new column from others

Example (go to MySQL Workbench)

```
SELECT
surname,
name,
age,
CONCAT(surname, ", ", name) AS complete_name,
age + 10 AS age_more
FROM social_network.user_table;
```

SELECT

CONCAT(column1, column2) AS new_column1
(column3 + column4) AS new_column2

Columns to retrieve

FROM

table_name

Table to query

WHERE

condition

Filter criteria

ORDER BY

column3 [ASC|DESC]

Sorting order

https://www.w3schools.com/sql/sql_operators.asp

Exercises

Using linkedin_jobs schema:

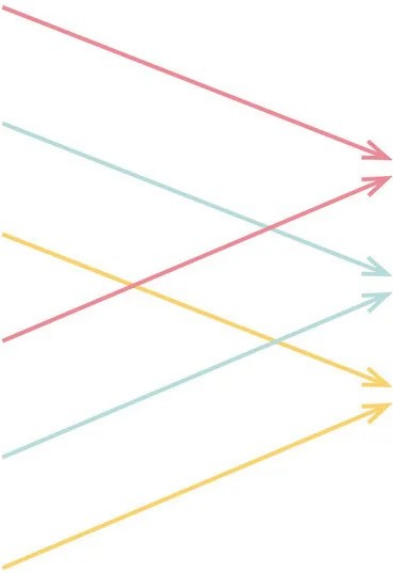
3) Show data of columns of company_id, name and create a column with complete address (address, city, state, country and zip_code) from table linkedin_jobs.companies_companies;



SQL – GROUP BY

Aggregate rows that have the same values into summary rows

title	genre	qty
book 1	adventure	4
book 2	fantasy	5
book 3	romance	2
book 4	adventure	3
book 5	fantasy	3
book 6	romance	1



genre	total
adventure	7
fantasy	8
romance	3

SQL – Aggregation Functions

Some of the aggregation function are:

Function	Description
COUNT(column)	Count not NULL values of the column within group
COUNT(*) or COUNT(1)	Count the number of rows within group.
COUNT(DISTINCT column)	Count the number of distinct values of the column within group
MIN(column), MAX(column)	Minimum and maximum of a column
AVG(column), SUM(column)	Average and sum of a column

<https://dev.mysql.com/doc/refman/8.4/en/aggregate-functions.html>

SQL – GROUP BY

Aggregate rows that have the same values into summary rows;

Example (go to MySQL Workbench)

```
SELECT
post_type,
COUNT(user) AS total_posts,
COUNT(DISTINCT user) AS total_distinct_users
FROM social_network.posts_table
GROUP BY post_type;
```

```
SELECT
AVG(age) AS average_age
FROM social_network.user_table
GROUP BY surname;
```

SELECT



column1, column2, .., AGG_FUNC(column3)

Columns to retrieve

FROM



table_name

Table to query

GROUP BY



column1, column2, ...

Group By Columns

Exercises

Using linkedin_jobs schema:

- 1) Count the the total of post per formatted_work_type in table postings
- 2) Calculate the average and maximum of company_size per country and state from table companies_companies



is used to combine rows from two or more tables, based on a related column between them.

ID	name	date_birth
1	Peter	14/05/1986
2	John	12/08/2001
4	Maria	4/09/1976

ID	UserID	class	grade
XDF	1	Mathematics	14
34FF	2	Arts	13
E45	3	Physics	16

Left

Right

Inner

s.ID	name	date_birth	g.ID	UserID	class	grade
1	Peter	14/05/1986	XDF	1	Mathematics	14
2	John	12/08/2001	34FF	2	Arts	13
4	Maria	4/09/1976	NULL	NULL	NULL	NULL

s.ID	name	date_birth	g.ID	UserID	class	grade
1	Peter	14/05/1986	1	XDF	Mathematics	14
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s.ID	name	date_birth	g.ID	UserID	class	grade
1	Peter	14/05/1986	1	XDF	Mathematics	14
2	John	12/08/2001	2	34FF	Arts	13

SQL – JOIN

is used to combine rows from two or more tables, based on a related column between them.

Student s

ID	name	date_birth
1	Peter	14/05/1986
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Grades g

ID	UserID	class	grade
XDF	1	Mathematics	14
34FF	2	Arts	13
E45	3	Physics	16
URFL	1	Music	18

Left

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Right

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2	John	12/08/2001	2	34FF	Arts	13

SQL - JOIN

is used to combine rows from two or more tables, based on a related column between them

Example (go to MySQL Workbench)

SELECT

```
o.orderNumber,  
o.status,  
od.quantityOrdered,  
od.priceEach,  
od.orderNumber
```

FROM classicmodels.Orders o

INNER JOIN classicmodels.OrderDetails od **ON** o.orderNumber=od.orderNumber;

SELECT

column1, column2, ...

Columns to retrieve

FROM

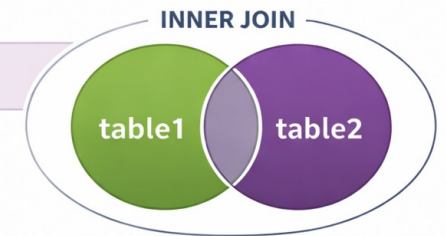
table1

First table

JOIN

table2

This query



ON

table1.column = table2.column

Join Condition

Exercises

Using linkedin_jobs schema:

- 1) Join table jobs_salaries and postings with a inner join using job_id column
- 2) Join table companies_companies and postings with a right join using company_id column. Select only columns job_id and title from postings table and from companies_companies table just the columns name and description

Quizz Time

It's Quizz time! Go to Virtual Campus and go to SQL – Quizz 1!

Good Luck



Make change happen

ACCREDITATIONS



MEMBERSHIPS



RANKINGS



SQL – SubQueries (using With)

a subquery can be defined as a query embedded within another query.

Using WITH	Not using WITH
WITH cte1 AS (SELECT a, b FROM table1), cte2 AS (SELECT c, d FROM table2) SELECT b, d FROM cte1 JOIN cte2 WHERE cte1.a = cte2.c;	SELECT b, d FROM (SELECT a, b FROM table1) cte1 JOIN (SELECT c, d FROM table2) cte2 WHERE cte1.a = cte2.c;

Example without WITH

```
SELECT o.orderNumber, o.status,od.quantityOrdered,od.priceEach,od.orderNumber
FROM classicmodels.Orders o
INNER JOIN (SELECT * FROM OrderDetails WHERE status = "Shipped") od ON
o.orderNumber=od.orderNumber;
```


SQL – SubQueries (using With)

a subquery can be defined as a query embedded within another query.

Using WITH	Not using WITH
WITH cte1 AS (SELECT a, b FROM table1), cte2 AS (SELECT c, d FROM table2) SELECT b, d FROM cte1 JOIN cte2 WHERE cte1.a = cte2.c;	SELECT b, d FROM (SELECT a, b FROM table1) cte1 JOIN (SELECT c, d FROM table2) cte2 WHERE cte1.a = cte2.c;

Example with WITH

```
WITH
  shipped_orders AS (SELECT * FROM OrderDetails WHERE status = "Shipped")

SELECT o.orderNumber, o.status,od.quantityOrdered,od.priceEach,od.orderNumber
FROM classicmodels.Orders o
INNER JOIN shipped_orders od ON o.orderNumber=od.orderNumber;
```

Exercises

- 1) Join table Products and ProductsLines, only with productLine with value "Ships", with a left join
- 2) Join table Payments and Customers, from USA country, with a left join using customerNumber column

Using the eleicoes database, create SQL query that respond the questions:

- 1) What is the code (column codigo) of the municipality (concelho) of Porto?
- 2) Number of votes (column votos from table votacoes) per party (partido)?
- 3) Number of votes per party in the municipality (concelho) of Porto?
 - table votacoes connect with table freguesias using column freguesia (votacoes) and column codigo (freguesias)
 - table freguesia connect with table concelho using column concelho (freguesia) and column codigo (concelho)
- 4) What are the names of the municipalities in the district (distrito) with the highest number of voters (column votantes from table participacoes)?
- 5) Top 10 parishes (freguesias) with the most votes (votacoes)

